Rogue AI Detection in the Wild

Findings, challenges and insights on detecting and preventing rogue AI agent activity

Whitepaper | 23 October 2026 | Companion to the methods report and the Python prototype

Data notice: the prototype described in this paper runs only on synthetic data generated by the project. No real platform, account or person was scraped or analyzed. Prototype results show that the pipeline works; they are not evidence of real-world detection accuracy.

1. Executive summary

AI agents that act through tools and APIs can be hijacked, misused or deliberately deployed for abuse. Detecting this from outside the systems an organization controls is hard, and the published evidence says no single technique is reliable. This paper summarizes what a review of public sources supports, shows how a small prototype puts those ideas together, and sets out what should be done to prevent rogue AI activity rather than only detect it.

- Layer the defenses. Unsupervised behavioral scoring, graph-based coordination analysis and tool-call monitoring each fail in different ways. Combining them, with human review, performed far better than the Isolation Forest alone in the prototype's synthetic test.
- Score behavior, not intent. Intent cannot be observed. Every flag is a lead for a person, and unsupervised evidence on its own should be capped at low severity until corroborated.
- Prevention beats detection. OWASP and MITRE ATLAS both frame agent harm around tool use, permissions and autonomy. Least privilege, approval for high-impact actions and complete logging reduce what a rogue agent can do and make misuse visible [4][5][7].
- Evasion is easy. The prototype's evasive variants dropped fused recall from 1.00 to 0.11 on synthetic data. Published work also shows that LLM-based detectors can be attacked [10].
- Collection is bounded by law and ethics. US case law has narrowed the CFAA, but contract terms, data protection law and platform rules still apply, so building on synthetic or pseudonymized data should be the default [14][15][16][17].

2. Problem and scope

There is no standard definition of a rogue AI agent. This work uses a working one: an automated, model-driven or model-assisted actor that pursues actions outside its operator's intended or authorized scope, or is operated to cause harm, on systems the detector's owner does not control. It covers two cases that look the same from outside: an agent hijacked through injected instructions, and an agent deliberately deployed for abuse.

The project brief names six components: data collection, anomaly detection, real-time monitoring with alerting, collaboration and intelligence sharing, an ethical and compliance framework including responsible disclosure, and a community engagement platform. The prototype implements each at small scale; the report covers the underlying methods and sources in detail.

3. What the evidence says

3.1 Bot and agent detection

Social bot detection is the most mature neighboring field. Botometer evolved from BotOrNot (2016) toward ensembles of specialized classifiers [8]. Yang et al. found that choosing a subset of training data generalized better across datasets than training on everything [9]. The TwiBot-22 benchmark is a graph with 92,932,326 nodes and 170,185,937 edges [11]. Most of this work targets accounts on one platform; an agent acting through APIs has no public profile to classify. Chan et al. propose agent identifiers, real-time monitoring and activity logs, all of which depend on the deployer's cooperation [12].

LLM-based detectors are an attack surface. Orenstein and Birman report accuracy drops of up to 48 percent from adversarial rewriting and prompt injection against LLM bot detectors, and about 86 percent accuracy for a multi-LLM ensemble under strong attack, on 2,000 TwiBot-20 accounts with three open-source models [10]. This is one group's result on one dataset.

3.2 Anomaly and coordination detection

Isolation Forest isolates anomalies with random splits and is a sensible first-pass detector for per-agent features [13]. Pacheco et al. build coordination networks from shared behavioral traces and link accounts only when their behavior is more similar than chance; they stress that coordination does not imply automation [6]. Meta's definition of coordinated inauthentic behavior is likewise behavior-based rather than content-based [18]. For agent-to-agent collusion, Motwani et al. show that steganographic coordination can evade monitoring and that simple mitigations may be insufficient [19].

3.3 Prompt injection and agent misuse

Greshake et al. demonstrated that instructions placed in retrieved data can steer LLM-integrated applications, with impacts including data theft and code execution [3]. OWASP lists Prompt Injection (LLM01) and Excessive Agency (LLM06) and names excessive functionality, permissions and autonomy as root causes [4]; its agentic list puts Agent Goal Hijack first [5]. NIST treats agent hijacking as a form of indirect prompt injection [20]. Anthropic reported in November 2025 that it disrupted an espionage campaign in which its coding tool performed much of the tactical work; this is the vendor's own account and was not independently verified here [21].

3.4 Standards and frameworks

Framework	Role in this work
MITRE ATLAS [1]	Shared vocabulary; alerts carry approximate technique tags (checked against the v2026.09 data file)
NIST AI RMF and AI 600-1 [2]	Governance functions (govern, map, measure, manage) and the generative AI profile
OWASP LLM and Agentic Top 10 [4][5]	Misuse patterns and mitigations
STIX 2.1 and TAXII 2.1 [22]	Structured sharing of indicators; the prototype exports STIX
ISO/IEC 29147 and CERT/CC CVD [23]	Template for a disclosure process
EU AI Act Article 73 [24]	Serious-incident reporting for high-risk AI providers; Annex III rules apply from 2 December 2027 after the Digital Omnibus

4. The prototype

The prototype is a Python application with a command line and a local web dashboard. It uses only synthetic events generated by the project; its network collector is intentionally disabled. It implements:

- A collector interface with a synthetic generator, a validated and PII-redacting importer for community-contributed files, and the disabled live stub.
- Per-agent feature extraction (timing, access, tool use, content and network features).
- Isolation Forest scoring with per-agent explanations, transparent rules keyed to technical gates (denied or out-of-scope requests, new external-destination tool use after reading external content, encoded payloads), and a coordination graph combining timing synchrony, rare-template content similarity and reciprocal messaging.
- A hash-chained alert log with severity, explanation and approximate ATLAS tags; a sliding-window monitor; flagged-case evidence packages; STIX 2.1 export; a community report form; and developer and disclosure documents.

The synthetic data plants three rogue roles (an access prober, a five-agent collusion ring and two injection-driven hijacks) among 53 benign agents, including look-alikes that should not be escalated: scheduled jobs, a heavy crawler, an orchestrator with workers, and routine users of an external tool. The roles are planted by the same author as the detectors, so the results below are a regression check, not a benchmark.

Synthetic run (seed 7, 62 agents, 9 rogue)	Precision	Recall	Notes
Fused pipeline, standard scenarios	1.00	1.00	No benign look-alike escalated above info or low
Isolation Forest alone, standard	0.62	0.56	Flags scheduled jobs and the heavy crawler; misses most of the collusion ring
Fused pipeline, stealth variant	1.00	0.11	1 of 9 found; slow probing, de-synchronized ring and a routine-tool hijack evade
Isolation Forest alone, stealth	0.25	0.22	Mostly false positives

Test status of the first version: 42 automated tests passed (Python 3.13.5, scikit-learn 1.9.1, numpy 2.5.4, networkx 3.7), covering schema and collectors, features, detectors, the pipeline on several seeds, alert-log tamper detection, the STIX export, report validation and the web endpoints. The documented install and run steps were also checked in a clean virtual environment.

Later versions (v3 to v5). The companion code has since grown into two projects. agent-watch is a localhost-only honeypot and log detector; rogue-ai-pipeline is the multi-layer scorer. v3 added a mirrored access stream, passive web detectors (asset starvation, unlinked jumps), score floors and a decoy MCP endpoint. v4 separated selective asset blocking from true starvation, added an ARIA canary and a three-boolean interaction beacon for browser-backed mimics, a two-signal corroboration gate, per-layer attribution of benign leads, an empirical link-graph fallback and an encoded-secret scrubber port. v5 adds a deterministic harness fingerprint (six archetypes, each with reasons and a low, medium or high confidence that describes how much of the signature was seen) and a static, escaped HTML replay page for human reviewers. Current test status: agent-watch 446 passed and 1 skipped, rogue-ai-pipeline 331 passed, pipeline type checks clean. These are regression checks on synthetic data.

What the prototype shows, and does not

- Shows: that independent evidence types can be fused into explained alerts, that benign periodic or orchestrated behavior can be handled explicitly, and that evasions are easy to plant and expected to work.
- Does not show: performance on real traffic, behavior under drift, or resistance to adaptive attackers. No such claim is made.

5. Challenges

Challenge	Why it matters
Intent and attribution	Behavior can be anomalous for benign reasons; a flag is not proof, and naming an operator wrongly causes real harm.
Evasion and adaptation	Adversarial rewriting, slow-and-low activity, and abuse of already-granted tools defeat simple detectors.
No external agent identity	Outside a deployer's own systems there is no agreed way to identify an AI agent or obtain its logs [12].
Labeled data	Little public labeled data exists for agent-to-agent misconduct; benchmarks such as TwiBot-22 are social-platform specific and gated [11].
Benign look-alikes	Schedulers, crawlers and orchestrators are unusual or synchronized too. Fan-out workloads triggered by one cause can look like coordination.
Detector drift and attack surface	LLM-based detectors can be attacked and can change as models are updated [10].
Legal and privacy limits	Collection must respect terms, data protection law and platform rules; EU platform data has a formal researcher route [14][15][16][17].
Human workload	Alert fatigue and the cost of review limit how aggressive thresholds can be.

6. Preventing rogue AI activity

Detection from outside is a backstop. The larger gains come from controls on the systems that host agents:

- Least privilege. Give each agent only the tools, data and permissions its task needs; avoid open-ended tools; run actions in the requesting user's context [4].
- Approval for high-impact actions. Require a human or independent check for deletion, external sends and privilege changes [4].
- Treat external content as untrusted. Retrieved pages, files and tool outputs can carry instructions; monitor for goal changes after ingestion [3][5].
- Identity and logging. Assign agent identifiers and keep tamper-evident activity logs so misuse can be attributed and investigated [12].
- Baseline and alert on tool-call sequences. Behavioral deviation (a new external-destination tool after reading external content) is more robust than content filtering.

- Incident process. Prepare detection, investigation, correction and reporting workflows, including obligations such as AI Act Article 73 where they apply [24].
- Share carefully. Exchange indicators in structured form with partners, after review and with personal data removed [22][23].

7. Ethics, legal constraints and responsible disclosure

Four sources set the boundaries most relevant to data collection. This is a research summary, not legal advice.

Source	Constraint	Design consequence
Van Buren v. United States (2021) [14]	CFAA 'exceeds authorized access' means entering areas one is barred from, not misusing allowed access	Define unauthorized access by technical gates; never bypass controls
hiQ v. LinkedIn (9th Cir. 2022) [15]	Upheld a preliminary injunction; CFAA likely does not reach public pages; no final merits ruling	Public-page collection has support in that circuit; other claims remain
Meta v. Bright Data (N.D. Cal. 2024) [16]	Contract claim failed on logged-off scraping of public data	Read each platform's terms; logged-in access is governed by them
EDPB Guidelines 03/2026 (draft) [17]	Scraping for generative AI training: minimize, consider synthetic data, exclude sites that oppose scraping	Scope is AI training by private entities; apply by analogy. Prefer synthetic or pseudonymized data

The Menlo Report's four principles (respect for persons, beneficence, justice, and respect for law and public interest) frame the project policy [25]. The project ships a collector compliance checklist and responsible disclosure guidelines adapted from coordinated vulnerability disclosure practice: report privately first, state confidence honestly, give the operator a route to dispute, and publish only aggregate lessons.

8. Recommendations

Near term

- Adopt the collector checklist and keep all development on synthetic or pseudonymized data until legal and ethics review is complete.
- Instrument agents you operate with identifiers, scoped credentials and tool-call logs, and feed them to the detectors.
- Extend the evaluation with adversarial and benign-look-alike cases, and report false positives and review workload alongside detection rates.

Medium term

- Build partnerships with security researchers and academic groups for labeled data and independent validation; use sanctioned data-access routes for platform data.
- Stand up a disclosure program with a published contact, dispute path and partner review before sharing indicators.
- Add drift monitoring and retraining on reviewed windows; test detectors against adaptive attackers.

Open research questions

- An accepted external identity for AI agents.
- Public labeled data for agent-to-agent misconduct.
- Whether communication monitoring can catch steganographic collusion.

9. Related work and comparative architecture

This section compares the companion code with work that already exists: a honeypot and log detector (decoy API documents, a decoy MCP endpoint, canary pages, a coordination board with a handoff token and nonce, a secret scrubber, passive detectors for asset starvation and unlinked jumps, an ARIA canary and a three-boolean interaction beacon, score floors, a harness fingerprint, a static replay page) plus the multi-layer pipeline. The comparison rests on a limited public web search on 23 October 2026, not a systematic review. Each source in the table was opened and read; the full table, dates and the items that could not be verified are in the project file docs/prior-art.md [45]. Wherever the table says “not found in my search”, that means exactly that: I did not find it documented in the pages I read, which is not evidence that the capability is absent. Drosera [27] overlaps the web side most closely.

Capability	Documented in prior art (what I read)	This stack (all numbers synthetic, same author)
Hidden-instruction canary pages	Palisade [26]; Drosera [27]; Thinkst [29]. Documented.	Yes. Contact with a canary is a lead, not proof of intent.
Per-session token / nonce round trip	Drosera: HMAC-signed ticket [27]. Others: not found in my search.	Handoff token plus nonce on the coordination board; a token replayed from another session or address is a separate signal (prior art: not found in my search).
Decoy MCP endpoint, tools/list then call	Sundew [28]; MCP Decoy [31]; HoneyMCP [32]; Thinkst [30]. Documented.	Yes; the harness fingerprint adds an mcp_tool_runner label.
Asset starvation	Drosera: no-assets behavior signal [27]. Others: not found in my search.	Yes. Since v4 it separates selective asset blocking from true starvation: a client that loads CSS or script, or sends coherent Sec-Fetch headers, is not flagged.
Move checked against the links the page actually offered	Not found in my search.	Yes (unlinked jump), with an empirical fallback from Referer transitions when the log does not record offered links. The fallback is circular within one batch.
Browser-backed mimics	FP-Agent and Fayolle et al. study client-side and multi-layer fingerprints [38][39]. Honeypot traps for this case: not found in my search.	ARIA canary plus a beacon of three booleans (no identifiers). Catches simulated mimics that fetch our script but never run it. JavaScript-disabled and beacon-blocking people produce no beacon: a known false positive.
Harness archetype labels	Model attribution from action traces [40]. Deterministic archetype labels for web agents: not found in my search.	Six archetypes plus unclassified, each with plain-language reasons; descriptive, never scored.

Capability	Documented in prior art (what I read)	This stack (all numbers synthetic, same author)
Downstream pipeline	Drosera documents noisy-OR fusion, log replay and a STIX report format [27]. A pipeline combining anomaly, rules, graph, audit chain and STIX: not found in my search.	Isolation Forest, rules, coordination graph, a two-signal corroboration gate, per-layer attribution, hash-chained audit log, STIX 2.1.
TLS and header fingerprints	JA3/JA4 [37]; Fayolle et al. [39]. Documented.	Not implemented (a parse-only field).
Real-world evidence	Palisade: about 8.1 million real attempts [26]; Li et al. [41].	None. Localhost and synthetic only.

Not novel. Prior art covers the hidden-instruction canary page [26][27], the signed per-session token and the nonce round-trip [27], the coordination board's request that an agent disclose its task, tools and identity [29] [26], the decoy OpenAPI and llms.txt documents [27][28], the decoy MCP endpoint with a tools/list-then-call signal [28][30][31], invisible-link traps [35], asset starvation as a signal [27], timing [26], and, on the pipeline side, the hash-chained log [43], Isolation Forest [13], the coordination graph [6] and STIX export [22]. Drosera [27], Sundew [28], MCP Decoy [31] and the Thinkst products [29][30] are the nearest matches, and our honeypot side should be described as a reimplementation of these ideas plus extensions, not a new technique.

Where this stack extends prior art (possibly new, unproven). (1) Selective blocking versus starvation: the asset-starvation signal is typed by what was skipped, which removed the false positive for people who block images (in the agent-watch simulation, 15 of 15 such clients were alerted in v3 and 0 of 15 in v4). (2) Link-graph and trap evidence for browser-backed mimics: each page move is scored against the links the honeypot actually served, with an empirical fallback for ordinary logs, and an ARIA canary plus a three-boolean beacon target clients that look like browsers. In the 135-client pipeline simulation (9 mimic cohorts over 5 seeds) the share producing a lead rose from 0 percent in v2 to 33 percent in v3 and 78 percent in v4 (105 of 135). The cohort that loads CSS and script and sends correct headers (mimic_assets) went from 0 of 15 to 15 of 15, but only because the simulated mimic fetches our script and never runs it; two evaders (a stealth browser and a header forger) are still missed, 0 of 15 each. (3) Coordination and harness evidence: a handoff token replayed from a different session or address, a capped claim loop, and a deterministic harness-archetype label with reasons. (4) Integration and honest reporting: the pseudonymized output feeds a separate pipeline with a two-signal corroboration gate and per-layer attribution of every lead. The search found each half in separate projects but no project doing both; finding no equivalent is not proof that none exists.

What the extensions cost. The beacon rule produces false positives on people whose script is blocked: 15 of 15 simulated beacon-blocking humans and 10 of 10 no-asset, no-Sec-Fetch humans received a weak lead. The corroboration gate removed 11 statistical-only benign leads (7 compliance scanners, 4 noisy-denied clients) in the standard scenario set and lowered the false-positive rate from 0.06 to 0.04, at the price of 2 injected probers and, in the all-evasive set, all recall that depended on Isolation Forest and the sequence model alone (core recall 0.24 to 0.00). Both trade-offs are reported in the project READMEs.

Honest evaluation boundaries. All figures are synthetic same-author regression checks, not benchmarks: the simulated clients, including each mimic, were written by the same author as the detectors, with the rules in mind. A honeypot or beacon contact shows contact, not intent. Every output is a lead for human review; nothing blocks or accuses anyone. Palisade reports about 8.1 million real attempts [26] and the Li et al. honeysites 26.4 million requests [41]; we have no real traffic, no real browser or agent was driven against the code, and no TLS

fingerprint layer, although measurement studies find multi-layer fingerprints strongly discriminative [38][39] and some agents bypassed every defense tested [39].

10. Limitations of this paper

- The sources were gathered by web search; a few primary documents were read in full (the EDPB draft guidelines, the arXiv paper on LLM bot detectors, and the ATLAS v2026.09 data file). Others rest on abstracts or summaries and should be checked before being cited onward.
- Several 2026 items are recent and may change, including the EDPB draft guidelines, ATLAS v2026.09 and the EU Digital Omnibus changes.
- Company reports (Anthropic, Meta) are self-reported.
- Prototype results are on synthetic data planted by the authors and do not measure real-world performance.
- The related-work search in section 9 was a limited public web search on 23 October 2026, not a systematic review; for GitHub projects the README was read, not the code, and none was run. Where a capability is marked “not found in my search” it may still exist. Several projects are weeks or months old and may change. The v3 to v5 figures come from simulated clients written by the same author as the detectors.

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